BAIO Learning System

I PUC BIOLOGY

# MID-TERM PREPARATION

## Score High in 2026

- Important Topics
- Diagrams to Practise
- Exam Tips



**Narendra Sir**
Biology Lecturer
Founder/CEO, BAIO Learning System

# How to use this video

- Pause on every topic and **write the answer** in your notebook
- Then check it with the slide and correct mistakes
- Draw every diagram **as given by Narendra Sir**
- Revise the key facts again just before the exam

CHAPTER

# 1

# The Living World

📝 4 marks in the paper · 3.8% weightage



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# Key facts for MCQs

Part A · 1 mark each · 2 Qs

- **Species** is the basic unit of classification
- Scientific name: genus with a capital letter, species with a small letter, in italics or underlined — *Mangifera indica* (mango)

# Binomial nomenclature

Part B · 2 marks

- Naming an organism with **two words**: generic name + specific epithet
- Example: *Homo sapiens*
- Proposed by **Carolus Linnaeus**

CHAPTER

# 2

# Biological Classification

📝 10 marks in the paper · 9.5% weightage



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# Key facts for MCQs

Part A · 1 mark each · 2 Qs

- Fungal cell wall is made of **chitin**
- Red rot of sugarcane is caused by a **fungus**, *Colletotrichum falcatum*

# Kingdom Monera

Part C · 3 marks

- Prokaryotic, unicellular; no membrane-bound organelles
- Cell wall of **peptidoglycan**
- Most abundant; found even in hot springs, deserts, snow, deep oceans
- Autotrophic (photo / chemosynthetic) or heterotrophic
- Reproduce mainly by fission
- Examples: *E. coli*, *Nostoc*, *Anabaena*, *Mycoplasma*

CHAPTER 2 · 10 MARKS

# Whittaker's five kingdoms (1969)

Part D · 5 marks

| Kingdom | Key features | Examples |
|---|---|---|
| Monera | Prokaryotic, unicellular, peptidoglycan wall | Bacteria, cyanobacteria |
| Protista | Unicellular eukaryotes, aquatic | *Chlamydomonas*, *Paramecium* |
| Fungi | Chitin wall, heterotrophic | *Mucor*, *Agaricus*, yeast |
| Plantae | Cellulose wall, autotrophic | Algae to flowering plants |
| Animalia | No cell wall, holozoic | Sponges to mammals |

**Criteria: cell structure, body organisation, nutrition, reproduction, phylogeny**

CHAPTER 2 · 10 MARKS

# Five kingdoms — merits and demerits

Part D · 5 marks

## Merits

- Prokaryotes separated from eukaryotes
- Unicellular eukaryotes grouped in Protista
- Fungi separated from plants
- Reflects phylogeny better than two kingdoms

## Demerits

- Protista is a mixed group (*Chlamydomonas* with *Paramecium*)
- Archaebacteria and eubacteria together in Monera
- Viruses, viroids and lichens not placed
- Photosynthetic cyanobacteria kept away from plants

CHAPTER

# 3

# Plant Kingdom

📝 10 marks in the paper · 9.5% weightage



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# Key facts for MCQs

Part A · 1 mark each · 3 Qs

- Bryophytes = **amphibians of the plant kingdom**
- *Cycas* is a **living fossil** among gymnosperms
- *Chlorella*: protein-rich alga used by space travellers

# Economic importance of algae

Part B · 2 marks

- Food: *Porphyra*, *Laminaria*, *Sargassum*; supplements *Chlorella*, *Spirulina*
- Algin (brown algae) and carrageen (red algae) used commercially
- Agar from *Gelidium* and *Gracilaria* — for culture media, ice-creams, jellies
- Algae fix about half of the total $CO_2$ on earth

CHAPTER 3 · 10 MARKS

# Classes of algae — identify a, b, c, d

Part D · 5 marks

| Class | Common name | Pigments | Stored food | Cell wall |
|---|---|---|---|---|
| Chlorophyceae | Green algae | Chl a, b | **c** | Cellulose, pectose |
| Phaeophyceae | Brown algae | Chl a, c, **b** | Mannitol, laminarin | Cellulose, **d** |
| Rhodophyceae | **a** | Chl a, d, phycoerythrin | Floridean starch | Cellulose, pectin, polysulphate esters |

Section VI: write a = ..., b = ..., c = ..., d = ... against each letter

# Answers: a, b, c, d

Part D · 5 marks

| Class | Common name | Pigments | Stored food | Cell wall |
|---|---|---|---|---|
| Chlorophyceae | Green algae | Chl a, b | **c = Starch** | Cellulose, pectose |
| Phaeophyceae | Brown algae | Chl a, c, **b = fucoxanthin** | Mannitol, laminarin | Cellulose, **d = algin** |
| Rhodophyceae | **a = Red algae** | Chl a, d, phycoerythrin | Floridean starch | Cellulose, pectin, polysulphate esters |

CHAPTER

# 4

# Animal Kingdom

📝 16 marks in the paper · 15.2% weightage



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# Key facts for MCQs

Part A · 1 mark each · 3 Qs

- **Water vascular system** — Echinodermata
- Mammals are **homoiothermous** (warm-blooded)
- Sponges belong to phylum **Porifera**

# Chordates and non-chordates

Part C · 3 marks

| Chordates | Non-chordates |
| --- | --- |
| Notochord present | Notochord absent |
| CNS dorsal, hollow, single | CNS ventral, solid, double |
| Pharyngeal gill slits present | Gill slits absent |
| Heart ventral | Heart dorsal (if present) |
| Post-anal tail present | Post-anal tail absent |

# Phylum Arthropoda

Part D · 5 marks

- Largest phylum — about two-thirds of all named species
- Bilateral, triploblastic, segmented, coelomate; chitinous exoskeleton
- **Jointed appendages**; body = head, thorax, abdomen
- Respiration: gills, book gills, book lungs, tracheae; **open circulation**
- Excretion by **Malpighian tubules**; antennae, compound eyes, statocysts
- Examples: *Apis*, *Bombyx*, *Laccifer*, *Anopheles*, *Locusta*, *Limulus*

# Identify the phylum

Part D · 5 marks

| Key characters | Phylum |
|---|---|
| Proboscis gland, collar and trunk | Hemichordata |
| Comb plates, bioluminescence | Ctenophora |
| Jointed appendages, Malpighian tubules | Arthropoda |
| Notochord, hollow nerve cord, gill slits | Chordata |
| Calcareous shell, head-foot-visceral mass, radula | Mollusca |

CHAPTER

# 5

# Morphology of Flowering Plants

📝 11 marks in the paper · 10.5% weightage



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# Key facts for MCQs

Part A · 1 mark each · 1 Q

- Edible part of mango: fleshy **mesocarp**
- Venation = arrangement of veins in the lamina Part B · 2 marks
- **Reticulate** — network of veinlets (dicots)
- **Parallel** — veins run parallel (monocots)

# Phyllotaxy

Part C · 3 marks

- Arrangement of leaves on the stem
- **Alternate** — one leaf per node: china rose, mustard, sunflower
- **Opposite** — a pair per node: *Calotropis*, guava
- **Whorled** — more than two per node: *Alstonia*

# Placentation

Part D · 5 marks

- Arrangement of ovules within the ovary
- **Marginal** — pea
- **Axile** — china rose, tomato, lemon
- **Parietal** — mustard, *Argemone* (false septum)
- **Free central** — *Dianthus*, *Primrose*
- **Basal** — sunflower, marigold

CHAPTER

# 6

# Anatomy of Flowering Plants

📝 5 marks in the paper · 4.8% weightage



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# Key facts for MCQs

Part A · 1 mark each · 2 Qs

- Root hairs are outgrowths of **epidermal cells**
- Stomata are regulated by bean-shaped **guard cells**

# Open and closed vascular bundles

Part C · 3 marks

| Open | Closed |
| --- | --- |
| Cambium present | Cambium absent |
| Secondary growth possible | No secondary growth |
| Dicot stems | Monocot stems |
| Bundle sheath usually absent | Sclerenchymatous bundle sheath |

CHAPTER

# 7

# Structural Organisation in Animals (Frog)

📝 4 marks in the paper · 3.8% weightage



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# Key facts for MCQs

Part A · 1 mark each · 1 Q

- Frogs are **cold-blooded (poikilotherms)** — body temperature changes with the surroundings
- To escape extreme heat and cold, frogs shelter in deep burrows
- **Aestivation** — summer sleep of the frog
- **Hibernation** — winter sleep of the frog

# Digestive system of frog

Part C · 3 marks

- Alimentary canal is **short** — frogs are carnivores
- Mouth → buccal cavity → pharynx → oesophagus → stomach → intestine → rectum → **cloaca**
- Food caught by the **bilobed tongue**; stomach digests with **HCl and gastric juice**
- **Bile** (liver, stored in gall bladder) and **pancreatic juice** reach the duodenum by the common bile duct
- Absorbed by **villi and microvilli**; waste leaves through the cloaca

CHAPTER 7 · 4 MARKS

# Diagram — Digestive system of frog

Part C · 3 marks

## ✏ Draw the diagram as given by Narendra Sir

**Labels to include:** Mouth · buccal cavity · tongue · pharynx · oesophagus · stomach · duodenum · intestine (ileum) · rectum · cloaca · cloacal aperture · liver · gall bladder · common bile duct · pancreas

CHAPTER

# 8

# Cell: The Unit of Life

📝 21 marks in the paper · 20% weightage



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# Key facts for MCQs

Part A · 1 mark each · 2 Qs

- Plant cell wall is made of **cellulose**
- Ribosomes are made in the **nucleolus**
- **Mesosomes**: infoldings of bacterial plasma membrane — help in wall formation, DNA replication, respiration, secretion Part B · 2 marks

# Plant cell and animal cell

Part B · 2 marks

| Plant cell | Animal cell |
| --- | --- |
| Cellulose cell wall | No cell wall |
| Plastids present | No plastids |
| One large central vacuole | Vacuoles small or absent |
| Centrioles usually absent | Centrioles present |

# Diagram — Plant cell

Diagram · draw with labels

## ✏ Draw the diagram as given by Narendra Sir

**Labels to include:** Cell wall · plasma membrane · cytoplasm · nucleus · nucleolus · chloroplast · mitochondrion · central vacuole · tonoplast · Golgi apparatus · ER · ribosomes · plasmodesmata

# Fluid mosaic model

Part D · 5 marks

- Proposed by **Singer and Nicolson (1972)**
- **Lipid bilayer**: polar heads outside, non-polar tails inside
- **Quasi-fluid** — lipids and proteins move sideways
- Proteins: **peripheral** (surface) and **integral** (buried)
- RBC membrane: about 52% protein, 40% lipid
- Selectively permeable — passive and active transport

# Diagram — Mitochondrion

Part D · 5 marks

## ✏ Draw the diagram as given by Narendra Sir

**Labels to include:** Outer membrane · inner membrane · cristae · intermembrane space · matrix · $F_o$–$F_1$ particles · circular DNA · 70S ribosomes

# Structures and their organelles

Part D · 5 marks

| Structure | Found in |
|---|---|
| Cisternae | Golgi apparatus |
| Cristae | Mitochondrion |
| Thylakoids | Chloroplast |
| Nucleolus | Nucleus |
| Tonoplast | Vacuole |

CHAPTER

# 9

# Biomolecules

📝 12 marks in the paper · 11.4% weightage



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# Key facts for MCQs

Part A · 1 mark each · 2 Qs

- **Purines:** adenine, guanine · **Pyrimidines:** cytosine, thymine, uracil
- Proteins are polymers of **amino acids**
- **Polysaccharides**: chains of monosaccharides joined by glycosidic bonds — cellulose, starch, glycogen, inulin, chitin Part B · 2 marks

# Structure of a nucleotide

Part C · 3 marks

- **Nitrogenous base** + **pentose sugar** + **phosphate**
- Base + sugar (N-glycosidic bond at 1′C) = **nucleoside**
- Nucleoside + phosphate (phosphoester bond at 5′C) = **nucleotide**
- Nucleotides join by 3′–5′ **phosphodiester bonds**

# Factors affecting enzyme activity

Part D · 5 marks

- **Temperature** — optimum; low = inactive (reversible), high = denatured
- **pH** — each enzyme has an optimum pH
- **Substrate concentration** — velocity rises, then reaches $V_{max}$; $K_m$ = [S] at ½ $V_{max}$
- **Activators and inhibitors** — malonate competitively inhibits succinic dehydrogenase

CHAPTER

# 10

# Cell Cycle and Cell Division

📝 12 marks in the paper · 11.4% weightage



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# Key facts for MCQs

Part A · 1 mark each · 2 Qs

- **Metaphase**: chromosomes at the equator, spindle fibres on kinetochores
- **Crossing over**: exchange between non-sister chromatids, by recombinase

  Part B · 2 marks
- Crossing over occurs in **pachytene** of Prophase I

# Mitosis and meiosis

Part C · 3 marks

| Mitosis | Meiosis |
| --- | --- |
| Somatic cells | Reproductive cells |
| One division, 2 cells | Two divisions, 4 cells |
| 2n → 2n | 2n → n |
| No crossing over | Crossing over in Prophase I |
| Identical daughter cells | Genetically different cells |

# Phases of the cell cycle

Part D · 5 marks

- **$G_1$** — cell grows; no DNA replication
- **S** — DNA replicates (2C → 4C)
- **$G_2$** — proteins made for mitosis
- **M phase** — karyokinesis (P, M, A, T) + cytokinesis
- **$G_0$** — quiescent stage; cells exit from $G_1$

CHAPTER 10 · 12 MARKS

# Diagram — Cell cycle

Diagram · draw with labels

## ✏ Draw the diagram as given by Narendra Sir

**Labels to include:** $G_1$ · S · $G_2$ · M phase · $G_0$ · interphase · direction arrows

# Diagrams to practise

1 Plant cell

2 Mitochondrion

3 Cell cycle

4 Digestive system of frog

# Narendra Sir's tips — how to study

- Read a topic, close the book, write it from memory
- Practise every diagram at least three times
- Learn examples and scientific names with each topic
- Revise differences as tables
- Use the **BAIO Learning System** app daily for MCQs and flashcards

# Narendra Sir's tips — in the exam hall

- Read the whole paper first; plan your choices
- Part A: only the **first answer** is evaluated
- Match the marks: 2 marks = 2 sentences, 5 marks = 200–250 words
- Write in points, underline key words, give examples
- **Unlabelled diagrams get no marks**
- Keep the last 10 minutes to check



# All the best!

## Work hard, stay confident, score high.

Narendra Sir

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Practise more on the BAIO Learning System app